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GAS TURBINE ENGINE INCLUDING STUB-TANDEM VARIABLE INLET GUIDE VANES

Abstract

Gas turbine engine including stub-tandem variable inlet guide vanes are disclosed. An example gas turbine engine includes a compressor, a variable inlet guide vane upstream of the compressor, and a part-span vane positioned between at least a portion of the variable inlet guide vane and the compressor in an axial direction defined by the gas turbine engine.

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Background/Summary

FIELD OF DISCLOSURE

[0001] This disclosure relates generally to gas turbine engines and, more particularly, to gas turbine engine including stub-tandem variable inlet guide vanes.

BACKGROUND

[0002] A gas turbine engine generally includes, in serial flow order, an inlet section, a compressor section, a combustion section, a turbine section, and an exhaust section. In operation, air enters the inlet section and flows to the compressor section where one or more axial compressors progressively compress the air until it reaches the combustion section, thereby creating combustion gases. The combustion gases flow from the combustion section through a hot gas path defined within the turbine section and then exit the turbine section via the exhaust section.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] FIG. 1 is a schematic cross-sectional view of an example engine for an aircraft in which examples disclosed herein can be implemented.

[0004] FIG. 2 illustrates first example variable inlet guide vanes that can be implemented in the engine of FIG. 1.

[0005] FIG. 3 illustrates second example variable inlet guide vanes that can be implemented in the engine of FIG. 1.

[0006] FIG. 4 illustrates third example variable inlet guide vanes that can be implemented in the engine of FIG. 1.

[0007] FIG. 5 illustrates an example actuation system to adjust a pitch of the variable inlet guide vanes of FIGS. 2, 3, and/or 4.

[0008] FIG. 6 illustrates another view of the example actuation system of FIG. 5.

[0009] FIG. 7 illustrates another example actuation system to adjust a pitch of the variable inlet guide vanes of FIG. 3.

[0010] FIG. 8A illustrates a first example cross-section of the variable inlet guide vanes of FIG. 3 in association with the third example actuation system of FIG. 7.

[0011] FIG. 8B illustrates a second example cross-section of the variable inlet guide vanes of FIG. 3 in association with the third example actuation system of FIG. 7.

[0012] FIG. 8C illustrates a third example cross-section of the variable inlet guide vanes of FIG. 3 in association with the third example actuation system of FIG. 7.

[0013] FIG. 8D illustrates a fourth example cross-section of the variable inlet guide vanes of FIG. 3 in association with the third example actuation system of FIG. 7.

[0014] FIG. 9 illustrates another example actuation system to adjust a pitch of the variable inlet guide vanes of FIGS. 3 and/or 4.

[0015] FIG. 10 illustrates an example cross-section of the actuation system of FIG. 9.

[0016] FIG. 11 illustrates another example actuation system to adjust a pitch of the variable inlet guide vanes of FIGS. 3 and/or 4.

[0017] FIG. 12 illustrates an example cross-section of the actuation system of FIG. 9.

[0018] FIG. 13 illustrates another example actuation system to adjust a pitch of the variable inlet guide vanes of FIGS. 2, 3, and/or 4.

[0019] FIG. 14 illustrates another example actuation system to adjust a pitch of the variable inlet guide vanes of FIGS. 2, 3, and/or 4.

[0020] In general, the same reference numbers will be used throughout the drawing(s) and accompanying written description to refer to the same or like parts. The figures are not necessarily to scale. Although the figures show layers and regions with clean lines and boundaries, some or all of these lines and/or boundaries may be idealized. In reality, the boundaries and/or lines may be

unobservable, blendable, and/or irregular.

DETAILED DESCRIPTION

[0021] “Including” and “comprising” (and all forms and tenses thereof) are used herein to be open ended terms. Thus, whenever a claim employs any form of “include” or “comprise” (e.g., comprises, includes, comprising, including, having, etc.) as a preamble or within a claim recitation of any kind, it is to be understood that additional elements, terms, etc., may be present without falling outside the scope of the corresponding claim or recitation. As used herein, when the phrase “at least” is used as the transition term in, for example, a preamble of a claim, it is open-ended in the same manner as the term “comprising” and “including” are open ended. The term “and/or” when used, for example, in a form such as A, B, and/or C refers to any combination or subset of A, B, C such as (1) A alone, (2) B alone, (3) C alone, (4) A with B, (5) A with C, (6) B with C, or (7) A with B and with C. As used herein in the context of describing structures, components, items, objects and/or things, the phrase “at least one of A and B” is intended to refer to implementations including any of (1) at least one A, (2) at least one B, or (3) at least one A and at least one B.

Similarly, as used herein in the context of describing structures, components, items, objects and/or things, the phrase “at least one of A or B” is intended to refer to implementations including any of (1) at least one A, (2) at least one B, or (3) at least one A and at least one B. As used herein in the context of describing the performance or execution of processes, instructions, actions, activities, etc., the phrase “at least one of A and B” is intended to refer to implementations including any of (1) at least one A, (2) at least one B, or (3) at least one A and at least one B. Similarly, as used herein in the context of describing the performance or execution of processes, instructions, actions, activities, etc., the phrase “at least one of A or B” is intended to refer to implementations including any of (1) at least one A, (2) at least one B, or (3) at least one A and at least one B.

[0022] As used herein, singular references (e.g., “a”, “an”, “first”, “second”, etc.) do not exclude a plurality. The term “a” or “an” object, as used herein, refers to one or more of that object. The terms “a” (or “an”), “one or more”, and “at least one” are used interchangeably herein.

Furthermore, although individually listed, a plurality of means, elements, or actions may be implemented by, e.g., the same entity or object. Additionally, although individual features may be included in different examples or claims, these may possibly be combined, and the inclusion in different examples or claims does not imply that a combination of features is not feasible and/or advantageous.

[0023] As used herein, unless otherwise stated, the term “above” describes the relationship of two parts relative to Earth. A first part is above a second part, if the second part has at least one part between Earth and the first part. Likewise, as used herein, a first part is “below” a second part when the first part is closer to the Earth than the second part. As noted above, a first part can be above or below a second part with one or more of: other parts therebetween, without other parts therebetween, with the first and second parts touching, or without the first and second parts being in direct contact with one another.

[0024] As used in this patent, stating that any part (e.g., a layer, film, area, region, or plate) is in any way on (e.g., positioned on, located on, disposed on, or formed on, etc.) another part, indicates that the referenced part is either in contact with the other part, or that the referenced part is above the other part with one or more intermediate part(s) located therebetween.

[0025] As used herein, connection references (e.g., attached, coupled, connected, and joined) may include intermediate members between the elements referenced by the connection reference and/or relative movement between those elements unless otherwise indicated. As such, connection references do not necessarily infer that two elements are directly connected and/or in fixed relation to each other. As used herein, stating that any part is in “contact” with another part is defined to mean that there is no intermediate part between the two parts.

[0026] Unless specifically stated otherwise, descriptors such as “first,” “second,” “third,” etc., are used herein without imputing or otherwise indicating any meaning of priority, physical order,

arrangement in a list, and/or ordering in any way, but are merely used as labels and/or arbitrary names to distinguish elements for ease of understanding the disclosed examples. In some examples, the descriptor “first” may be used to refer to an element in the detailed description, while the same element may be referred to in a claim with a different descriptor such as “second” or “third.” In such instances, it should be understood that such descriptors are used merely for identifying those elements distinctly within the context of the discussion (e.g., within a claim) in which the elements might, for example, otherwise share a same name.

[0027] As used herein, “approximately” and “about” modify their subjects/values to recognize the potential presence of variations that occur in real world applications. For example, “approximately” and “about” may modify dimensions that may not be exact due to manufacturing tolerances and/or other real world imperfections as will be understood by persons of ordinary skill in the art. For example, “approximately” and “about” may indicate such dimensions may be within a tolerance range of $\pm 10\%$ unless otherwise specified herein.

[0028] Gas turbine engines include a fan section proximate to an intake of an engine. The fan section includes a plurality of circumferentially spaced apart fan blades. A rotating portion of the fan section including the fan blades is rotatably coupled to a low pressure (LP) compressor (e.g., booster) via an LP shaft. In some examples, the LP shaft includes an LP shaft portion and a fan shaft portion. To facilitate channeling the airflow from the fan section into the LP compressor, some known gas turbine engines include a plurality of circumferentially spaced apart inlet guide vanes.

[0029] Variable inlet guide vanes have an angular orientation (e.g., a pitch or stagger) that can be adjusted to obtain flow properties that improve engine performance (e.g., increase thrust, increase a pressure increase produced by (e.g., a pressure ratio of) the compressor, etc.). For example, a variable inlet guide vane can close (e.g., rotate in a first direction that causes the trailing edge of the vane to point further in an engine spool rotation direction than in a nominal position, rotate in a first direction that increases a difference between circumferential positions of a leading edge of the inlet guide vane and a trailing edge of the inlet guide vane). In this way, the air discharged from the vane passage carries a velocity component tangential to the engine spool rotation direction. This can be referred to as an increase in swirl (e.g., flow angle as measured in the circumferential direction relative to the engine axis, flow angle in the direction of spool rotation), which causes the rotor blade immediately downstream of the vane to encounter a different flow velocity (e.g., speed and angle) in a rotating frame of reference of the blade compared to when the variable inlet guide vane is in the nominal position. As a result of the variable inlet guide vane closing, the rotor blade typically experiences a reduced relative flow velocity and a reduced incidence compared to when the variable inlet guide vane is in the nominal position. As used herein, incidence refers to the angle between (i) a direction in which the working fluid (e.g., air) is flowing and (ii) an orientation of a leading edge of a blade or vane that the working fluid contacts. The reduced relative flow velocity and incidence causes the blade to perform less work. This, in turn, causes the blade to draw less flow. The variable inlet guide vane can open (e.g., rotate in a second direction opposite the first direction, rotate in a direction that causes the trailing edge of the vane to point further away from the engine spool rotation direction than in the nominal position) to provide an opposite aerodynamic effect to that of the variable inlet guide vane closing.

[0030] In single or multi-stage compressors, nonsynchronous vibration refers to a blade or vane vibration that is not an integer multiple of the engine spool rotational speed. Certain types of nonsynchronous vibration are due to the formation of discrete aerodynamic instabilities that rotate in the same direction as the rotor but at a lower speed. These instabilities change the unsteady aerodynamic loads experienced by the blades due to the difference in rotational speed between the blade and the instabilities, and that difference may coincide with a natural vibrational mode of the rotor structure. Nonsynchronous vibration can cause damage to the blades. The formation of these instabilities is related to the aerodynamic operating point approaching, but not reaching, the point

of flowfield breakdown or stall.

[0031] High-pressure-ratio multistage compressors implement variable vane systems due to natural aerodynamic mismatch effects that occur during operation away from the aerodynamic design point (ADP). At a given rotational speed, each stage (e.g., rotor-stator pair) of the compressor can operate down to a minimum effective downstream throttle setting before encountering a stability limit, at which point the flowfield can no longer be sustained (e.g., at which point a stall would occur). This effective throttle setting is equivalent to the flow drawn by the stage immediately downstream. At a rotational speed below the ADP speed, each of the stages in the compressor naturally draws a lower flow than nominal. Accordingly, the furthest downstream stages draw a lower flow, which closes the effective throttle of the middle stages (e.g., increases the pressure ratio across the middle stages). This effective throttle closure forces the middle stages to operate closer to their stability limits and causes the middle stages to draw even less flow. The variable vanes in the middle stages may close to relieve the increased flow incidence, diffusion, and work experienced by the middle stages. However, the closure of the variable vanes causes the middle stages to draw even less flow, which in turn causes the upstream stages to experience an even greater effective throttle closure than the middle stages. As such, the front variable vanes (e.g., the inlet variable guide vane and one or more downstream variable vanes at a front of the compressor) close even further than the variable vanes of the middle stages to offset this impact to the stability margin (e.g., an amount of throttle closure relative to the current position at which the compressor would exceed a corresponding stability limit).

[0032] Further, the 2-dimensional (2D) effects occur spanwise and, thus, are observed in both the axial direction and the radial direction. The upstream rotor includes a lower radius ratio (e.g., ratio of hub radius to tip radius) than the downstream rotor, and the lower radius ratio creates a greater difference in an axial velocity profile from a hub of the blade to a tip of the blade (e.g., a >40% difference between hub and tip axial velocity with a rotor inlet radius ratio of 0.4). As the inlet guide vane rotates further (e.g., in a direction that increases a difference between circumferential positions of a leading edge of the inlet guide vane and a trailing edge of the inlet guide vane) to increase swirl, the difference in the axial velocity profile from the hub to the tip continues to increase. With a high radius ratio blade, a flow incidence (e.g., tip incidence)—which correlates with (e.g., is directly related to) nonsynchronous vibration and general operability risk—decreases as the variable inlet guide vane closes. However, the increased difference in the axial velocity profile of an upstream blade having a lower radius ratio causes the axial velocity of the airflow to encounter a significant decrease. Accordingly, while an increase in swirl can help reduce the flow incidence, a reduction in axial velocity can also result from the increase in swirl, which can remove or reduce a benefit provided by the increase in swirl and may result in nonsynchronous vibration. As used herein, tip incidence or flow incidence refers to the angle between a direction in which fluid (e.g., air) is flowing relative to an orientation of a surface at a tip of an airfoil that the fluid contacts. Tip incidence can be calculated based on an angle between the relative velocity vector and an axial or meridional direction in a velocity triangle.

[0033] Examples disclosed herein provide stub-tandem variable inlet guide vanes that control swirl distribution to reduce the flow incidence and nonsynchronous vibration encountered by airfoils in a compressor.

[0034] In some examples, the stub-tandem variable inlet guide vanes include a first variable inlet guide vane that spans from an inner radial edge of a flow path defined at an inlet of a compressor to an outer radial edge of the flow path. That is, the first variable inlet guide vane spans a radial distance of the flow path (e.g., a first variable pitch airfoil, a full-span vane, etc.). The stub-tandem variable inlet guide vanes also include a second variable inlet guide vane at least partially downstream of the first variable inlet guide vane. The second variable inlet guide vane spans only a portion of the radial distance of the flow path (e.g., a second variable pitch airfoil, a part-span vane, a variable part-span vane, etc.). Specifically, the second variable inlet guide vane includes (i) an

outer radial edge at the outer radial edge of the flow path and (ii) an inner radial edge positioned in a mid-radial portion of the flow path (e.g., further from the inner radial edge of the flow path than an inner radial edge of the first variable inlet guide vane). As such, the first and second variable inlet guide vanes provide a second degree of freedom to control a tip incidence encountered by the downstream rotors and reduce nonsynchronous vibrations that the rotors encounter. Specifically, typical variable inlet guide vanes that span the full radial-length of the inlet provide a single degree of freedom to a spanwise swirl distribution. Advantageously, in examples disclosed herein, the first inlet guide vane provides a first degree of freedom that adjusts a swirl distribution across at least a portion of a radial length of the flow path, and the second (e.g., stub, part-span) variable inlet guide vane provides a second degree of freedom that can further adjust a swirl distribution near an outer radial boundary of the flow path.

[0035] Referring now to the drawings, FIG. 1 is a schematic cross-sectional view of an example turbo engine **100** that can incorporate various examples disclosed herein. The example turbo engine **100** can be implemented on an aircraft and therefore referred to as an aircraft engine. In this example, the turbo engine **100** is a turbofan-type of engine. However, the principles of the present disclosure are also applicable to other types of engines, such as turboprop engines and engines without a nacelle, such as unducted fan (UDF) engines (sometimes referred to as propfans). Further, the example principles disclosed herein can be implemented on other types of engines, such as non-aircraft engines.

[0036] As shown in FIG. 1, the turbo engine **100** includes an outer bypass duct **102** (which may also be referred to as a nacelle, fan duct, or outer casing), a gas turbine engine **104** (which may also be referred to as a core turbine engine), and a fan section **106**. The gas turbine engine **104** and the fan section **106** are disposed at least partially in the outer bypass duct **102**. The gas turbine engine **104** is disposed downstream from the fan section **106** and drives the fan section **106** to produce forward thrust.

[0037] As shown in FIG. 1, the turbo engine **100** and/or the gas turbine engine **104** define a longitudinal or axial centerline axis **108** extending therethrough for reference. FIG. 1 also includes an annotated directional diagram with reference to an axial direction A, a radial direction R, and a circumferential direction C. In general, as used herein, the axial direction A is a direction that extends generally parallel to the centerline axis **108**, the radial direction R is a direction that extends orthogonally outward from or inward toward the centerline axis **108**, and the circumferential direction C is a direction that extends concentrically around the centerline axis **108**. Further, as used herein, the term “forward” refers to a direction along the centerline axis **108** in the direction of movement of the turbo engine **100**, such as to the left in FIG. 1, while the term “rearward” refers to a direction along the centerline axis **108** in the opposite direction, such as to the right in FIG. 1.

[0038] The gas turbine engine **104** includes a substantially tubular outer casing **110** (which may also be referred to as a mid-casing) that defines an annular inlet **112**. The outer casing **110** of the gas turbine engine **104** can be formed from a single casing or multiple casings. The outer casing **110** encloses, in serial flow relationship, a compressor section having a booster or low pressure compressor **114** (“LP compressor **114**”) and a high pressure compressor **116** (“HP compressor **116**”), a combustion section **118**, a turbine section having a high pressure turbine **120** (“HP turbine **120**”) and a low pressure turbine **122** (“LP turbine **122**”), and an exhaust section **124**.

[0039] The gas turbine engine **104** includes a high pressure shaft **126** (“HP shaft **126**”) that drivingly couples the HP turbine **120** and the HP compressor **116**. The gas turbine engine **104** also includes a low pressure shaft **128** (“LP shaft **128**”) that drivingly couples the LP turbine **122** and the LP compressor **114**. The LP shaft **128** also couples to a fan shaft **130**.

[0040] The fan section **106** includes a plurality of fan blades **132** that are coupled to and extend radially outward from the fan shaft **130**. In some examples, the LP shaft **128** may couple directly to the fan shaft **130** (i.e., a direct-drive configuration). In alternative configurations, the LP shaft **128**

may couple to the fan shaft **130** via a reduction gear **134** (i.e., an indirect-drive or geared-drive configuration). While in this example the gas turbine engine **104** includes two compressors and two turbines, in other examples, the gas turbine engine **104** may only include one compressor and one turbine. Further, in other examples, the gas turbine engine **104** can include more than two compressors and turbines. In such examples, the gas turbine engine **104** may include more than two drive shafts or spools.

[0041] As illustrated in FIG. 1, during operation of the turbo engine **100**, air **136** enters an inlet portion **138** of the turbo engine **100**. The air **136** is accelerated by the fan blades **132**. A first portion **140** of the air **136** flows into a bypass airflow passage **142**, while a second portion **144** of the air **136** flows into the inlet **112** of the gas turbine engine **104** (and, thus, into the LP compressor **114**). Inlet guide vanes **145** are positioned proximate to the inlet **112** to help guide the flow of the second portion **144** of the air **136** to control downstream tip incidence and enhance engine performance. The inlet guide vanes **145** are stub-tandem (e.g., full span-part span, father-son) variable inlet guide vanes that reduce non-synchronous vibration, as discussed in further detail below. Downstream of the inlet guide vanes **145**, one or more sequential stages of LP compressor stator vanes **146** and LP compressor rotor blades **148** coupled to the LP shaft **128** progressively compress the second portion **144** of the air **136** flowing through the LP compressor **114** en route to the HP compressor **116**. Next, one or more sequential stages of HP compressor stator vanes **150** and HP compressor rotor blades **152** coupled to the HP shaft **126** further compress the second portion **144** of the air **136** flowing through the HP compressor **116**. This provides compressed air **154** to the combustion section **118** where it mixes with fuel and burns to provide combustion gases **156**. Fuel is injected into the combustion section **118** by one or more nozzles **157**. The turbo engine **100** includes a fuel system to provide pressurized fuel through the nozzles **157** to the combustion section **118** of the gas turbine engine **104**. Example fuel systems are disclosed in further detail herein.

[0042] The combustion gases **156** flow through the HP turbine **120** where one or more sequential stages of HP turbine stator vanes **158** and HP turbine rotor blades **160** coupled to the HP shaft **126** extract a first portion of kinetic and/or thermal energy. This energy extraction supports operation of the HP compressor **116**. The combustion gases **156** then flow through the LP turbine **122** where one or more sequential stages of LP turbine stator vanes **162** and LP turbine rotor blades **164** coupled to the LP shaft **128** extract a second portion of thermal and/or kinetic energy therefrom. This energy extraction causes the LP shaft **128** to rotate, which supports operation of the LP compressor **114** and/or rotation of the fan shaft **130**. The combustion gases **156** then exit the gas turbine engine **104** through the exhaust section **124** thereof. The combustion gases **156** mix with the first portion **140** of the air **136** from the bypass airflow passage **142**. The combined gases exit an exhaust nozzle **170** (e.g., a converging/diverging nozzle) of the bypass airflow passage **142** to produce propulsive thrust.

[0043] FIG. 2 illustrates first stub-tandem variable inlet guide vanes **200** that can implement the inlet guide vanes **145** of FIG. 1. The stub-tandem variable inlet guide vanes **200** of FIG. 2 include a first variable inlet guide vane **202** (e.g., a first part-span vane, a first variable pitch airfoil) and a second variable inlet guide vane **204** (e.g., a stub, a second part-span vane, a second variable pitch airfoil) that encounter the second portion **144** (FIG. 1) of the air **136** (FIG. 1) between the inlet **112** (FIG. 1) and the LP compressor **114** (FIG. 1). The first variable inlet guide vane **202** includes a first leading edge **206**, a first trailing edge **208**, a first inner radial edge **210**, a first outer radial edge **212**, and a first shaft **214**. The second variable inlet guide vane **204** includes a second leading edge **216**, a second trailing edge **218**, a second inner radial edge **220**, a second outer radial edge **222**, and a second shaft **224**.

[0044] In FIG. 2, the second portion **144** of the air **136** is in a flow path **226** defined between an inner radial surface **228** and an outer radial surface **230**. For example, the inner radial surface **228** can be at least partially defined by the LP fan shaft **128** to which the LP compressor rotor blades **148** (FIG. 1) connect downstream of the stub-tandem variable inlet guide vanes **200**. The outer

radial surface **230** can be defined by an inner radial surface of the outer casing **110** (FIG. 1). [0045] In FIG. 2, the second variable inlet guide vane **204** is supported at only one end (e.g., the second outer radial edge **222**) while the other end (e.g., the second inner radial edge **220**) is suspended in the flow path **226** between the inner radial surface **228** and the outer radial surface **230**. Specifically, the second variable inlet guide vane **204** is supported via the second shaft **224** connected to the second outer radial edge **222**. As such, the second variable inlet guide vane **204** is cantilevered. In FIG. 2, the first shaft **214** connects to both the first inner radial edge **210** and the first outer radial edge **212** and extends past both the inner radial surface **228** and the outer radial surface **230** to couple to a supporting structure (e.g., the LP fan shaft **128**, the outer casing **110** (FIG. 1), an actuator, etc.). Accordingly, the first variable inlet guide vane **202** is supported at both the first inner radial edge **210** and the first outer radial edge **212**. As such, the first variable inlet guide vane **202** is not cantilevered.

[0046] In FIG. 2, the first leading edge **206** of the first variable inlet guide vane **202** is aligned with the second leading edge **216** of the second variable inlet guide vane **204** in the axial direction A (e.g., in a same axial position). Similarly, the first trailing edge **208** of the first variable inlet guide vane **202** is aligned with the second trailing edge **218** of the second variable inlet guide vane **204** in the axial direction A (e.g., in a same axial position). In some other examples, the leading edges **206**, **216** and/or the trailing edges **208**, **218** are offset in the axial direction A, as discussed further in association with FIGS. 3 and 4.

[0047] In FIG. 2, the first inner radial edge **210** of the first variable inlet guide vane **202** is positioned at (e.g., is contiguous with) the inner radial surface **228**. The second outer radial edge **222** of the second variable inlet guide vane **204** is positioned at (e.g., is defined at, is contiguous with) the outer radial surface **230**. The first outer radial edge **212** of the first variable inlet guide vane **202** and the second inner radial edge **220** of the second variable inlet guide vane **204** are positioned in a mid-radial portion of the flow path **226** between the inner radial surface **228** and the outer radial surface **230**. In some examples, the first variable inlet guide vane **202** spans at least 50% of a radial height of the flow path **226** (e.g., a distance between the inner radial surface **228** and the outer radial surface **230**). For example, the first variable inlet guide vane **202** can span between 65% and 85% of the radial height of the flow path. In some examples, the first variable inlet guide vane **202** spans up to 90% of the radial height of the flow path. In this example, the second inner radial edge **220** is positioned between the first outer radial edge **212** and the outer radial surface **230**. As such, the first variable inlet guide vane **202** occupies a separate radial portion of the flow path **226** than the second variable inlet guide vane **204**. In some examples, the second variable inlet guide vane **204** spans from the outer radial surface **230** and occupies approximately 50% to approximately 10% of the radial height of the flow path **226**. For example, the second variable inlet guide vane **204** can extend to the outer radial surface **230** and be separated from the inner radial surface **228** in the radial direction by between 50% and 90% of the radial height of the flow path **226**.

[0048] In some examples, the first shaft **214** and/or the second shaft **224** are coupled to one or more actuators, which adjust a pitch of the first variable inlet guide vane **202** along a first trajectory **232** and the second variable inlet guide vane **204** along a second trajectory **234**. As a result, adjustments to the pitches of the inlet guide vanes **202**, **204** can adjust the tip incidence encountered by downstream rotor blades (e.g., the LP compressor rotor blades **148** (FIG. 1), the HP compressor rotor blades **152** of FIG. 1) and, thus, control (e.g., reduce) nonsynchronous vibration encountered by the rotor blades. In this example, the first variable inlet guide vane **202** and the second variable inlet guide vane **204** share a same rotational axis **236**. In some examples, the first variable inlet guide vane **202** and the second variable inlet guide vane **204** are actuated together in opposite directions or in different proportions to adjust the tip incidence and, thus, control (e.g., reduce) nonsynchronous vibration encountered by the compressor rotor blades. In some other examples, the first variable inlet guide vane **202** and the second variable inlet guide vane **204** are independently

actuated. In some examples, the first outer radial edge **212** is positioned as close to the second inner radial edge **220** as manufacturing tolerances permit without causing the edges **212**, **220** to rub as the vanes **202**, **204** rotate. For example, the first outer radial edge **212** can be separated from the second inner radial edge **220** by a distance from 0.005 inches (in.) to 0.050 (in.). Example actuation mechanisms associated with the stub-tandem variable inlet guide vanes **200** of FIG. 2 are discussed further in association with FIGS. 5, 6, 13, and/or 14.

[0049] FIG. 3 illustrates second stub-tandem variable inlet guide vanes **300** that can implement the inlet guide vanes **145** of FIG. 1. The stub-tandem variable inlet guide vanes **300** of FIG. 3 include a first variable inlet guide vane **302** (e.g., a full-span vane, a first variable pitch airfoil) and a second variable inlet guide vane **304** (e.g., a stub, a part-span vane, a second variable pitch airfoil) that encounter the second portion **144** (FIG. 1) of the air **136** (FIG. 1) between the inlet **112** (FIG. 1) and the LP compressor **114** (FIG. 1). The first variable inlet guide vane **302** includes a first leading edge **306**, a first trailing edge **308**, a second trailing edge **310**, a first inner radial edge **312**, a first outer radial edge **314**, a second outer radial edge **316**, and a first shaft **318**. More particularly, an inner radial portion **320** of the first variable inlet guide vane **302** includes the first inner radial edge **312**, the first trailing edge **308**, and the first outer radial edge **314**. The first shaft **318** connects to the first inner radial edge **312** and the second outer radial edge **316** and extends past the inner radial surface **228** and the outer radial surface **230** to couple to a supporting structure, (e.g., the LP fan shaft **128** (FIG. 1), the outer casing **110** (FIG. 1), an actuator, etc.). As such, the first variable inlet guide vane **302** is not cantilevered. The second variable inlet guide vane **304** includes a second leading edge **324**, a third trailing edge **326**, a second inner radial edge **328**, a third outer radial edge **330**, and a second shaft **332**. The second inner radial edge **328** is suspended in the flow path **226** between the inner radial surface **228** and the outer radial surface **230**. Accordingly, the second variable inlet guide vane **304** is cantilevered.

[0050] In FIG. 3, an outer radial portion **322** of the first variable inlet guide vane **302** is defined outward of the first outer radial edge **314** in the radial direction R and includes the second trailing edge **310** and the second outer radial edge **316**. Further, a leading portion **334** (e.g., an upstream portion) of the first variable inlet guide vane **302** is defined between the first leading edge **306** and the second trailing edge **310** in the axial direction A. A trailing portion **336** (e.g., a downstream portion) of the first variable inlet guide vane **302** is defined between the second trailing edge **310** and the first trailing edge **308** in the axial direction A.

[0051] In FIG. 3, the first inner radial edge **312** of the first variable inlet guide vane **302** is positioned at (e.g., is defined at, is contiguous with) the inner radial surface **228** of the flow path **226**. The second outer radial edge **316** of the first variable inlet guide vane **302** and the third outer radial edge **330** of the second variable inlet guide vane **304** are positioned at (e.g., are defined at, are contiguous with) the outer radial surface **230** of the flow path **226**. As such, the leading portion **334** of the first variable inlet guide vane **302** spans from the inner radial surface **228** to the outer radial surface **230** (e.g., spans a height of the flow path **226**). The second inner radial edge **328** is positioned between the first outer radial edge **314** and the third outer radial edge **330** in the radial direction R. As such, the trailing portion **336** of the first variable inlet guide vane **302** spans only a portion of the height of the flow path **226** between the first inner radial edge **312** and the first outer radial edge **314**. In some examples, the second variable inlet guide vane **304** spans from the outer radial surface **230** and occupies from approximately 50% to approximately 10% of the radial height of the flow path **226**. Moreover, the second variable inlet guide vane **304** is aligned with the outer radial portion **322** of the first variable inlet guide vane **302** in the radial direction R. The second variable inlet guide vane **304** is positioned downstream of the outer radial portion **322**. More particularly, the second leading edge **324** is positioned between the first trailing edge **308** and the second trailing edge **310** in the axial direction A. Additionally, the third trailing edge **326** is positioned aft of the first trailing edge **308**.

[0052] In some examples, the first shaft **318** and/or the second shaft **332** are coupled to one or more

actuators, which adjust a pitch of the first variable inlet guide vane **302** along a first trajectory **342** and the second variable inlet guide vane **304** along a second trajectory **344**. As a result, adjustments to the pitches of the inlet guide vanes **302**, **304** can adjust the tip incidence encountered by downstream rotor blades (e.g., the LP compressor rotor blades **148**, the HP compressor rotor blades **152** of FIG. **1**) and, thus, control (e.g., reduce) nonsynchronous vibration encountered by the rotor blades. In this example, the first shaft **318** defines a first rotational axis **338**, and the second shaft **332** defines a second rotational axis **340** downstream of the first rotational axis **338**. In some examples, the second rotational axis **340** is positioned downstream of the first trailing edge **308**. In some other examples, the second rotational axis **340** is positioned upstream of the first trailing edge **308** and overlaps the trailing portion **336** of the first variable inlet guide vane **302**, as shown in the illustrated example of FIG. **7**. In some such examples, the third trailing edge **326** has a same axial position as the first trailing edge **308**. In some other such examples, the third trailing edge **326** is still downstream of the first trailing edge **308**.

[0053] In some examples, the first variable inlet guide vane **302** and the second variable inlet guide vane **304** are actuated together in opposite directions or in different proportions to adjust the tip incidence and, thus, control (e.g., reduce) nonsynchronous vibration encountered by the compressor rotor blades. In some other examples, the first variable inlet guide vane **302** and the second variable inlet guide vane **304** are independently actuated. Example actuation mechanisms associated with the stub-tandem variable inlet guide vanes **300** of FIG. **3** are discussed further in association with FIGS. **5**, **6**, **7**, **8A-8D**, **9**, **10**, **11**, **12**, **13**, and/or **14**.

[0054] In some examples, the second leading edge **324** is positioned as close to the second trailing edge **310** in the axial direction **A** as manufacturing tolerances permit without causing the vanes **302**, **304** to rub when rotating. In some other examples, the second leading edge **324** contacts and/or overlaps the first variable inlet guide vane **302** in the axial direction **A**. In some examples, the second inner radial edge **328** is positioned as close to the first outer radial edge **314** as manufacturing tolerances permit without causing the vanes **302**, **304** to rub when rotating.

[0055] FIG. **4** illustrates third example stub-tandem variable inlet guide vanes **400** that can implement the inlet guide vanes **145** of FIG. **1**. The stub-tandem variable inlet guide vanes **400** of FIG. **4** include a first variable inlet guide vane **402** (e.g., a full-span vane, a first variable pitch airfoil) and a second variable inlet guide vane **404** (e.g., a part-span vane, a second variable pitch airfoil) that encounter the second portion **144** (FIG. **1**) of the air **136** (FIG. **1**) between the inlet **112** (FIG. **1**) and the LP compressor **114** (FIG. **1**). The first variable inlet guide vane **402** includes a first leading edge **406**, a first trailing edge **408**, a first inner radial edge **410**, a first outer radial edge **412**, and a first shaft **414**. The first shaft **414** connects to the first inner radial edge **410** and the first outer radial edge **412** and extends past the inner radial surface **228** and the outer radial surface **230** to couple to a supporting structure, (e.g., the LP fan shaft **128** (FIG. **1**), the outer casing **110** (FIG. **1**), an actuator, etc.). The second variable inlet guide vane **404** includes a second leading edge **416**, a second trailing edge **418**, a second inner radial edge **420**, a second outer radial edge **422**, and a second shaft **424**. The second inner radial edge **420** is suspended in the flow path **226** between the inner radial surface **228** and the outer radial surface **230**. Accordingly, the second variable inlet guide vane **404** is cantilevered.

[0056] In FIG. **4**, the first inner radial edge **410** of the first variable inlet guide vane **402** is positioned at (e.g., is defined at, is contiguous with) the inner radial surface **228** of the flow path **226**. The first outer radial edge **412** and the second outer radial edge **422** are positioned at (e.g., are defined at, are contiguous with) the outer radial surface **230** of the flow path **226**. As such, the first variable inlet guide vane **402** spans a height of the flow path **226** from the inner radial surface **228** to the outer radial surface **230**. In FIG. **4**, the second inner radial edge **420** is defined at a mid-radial portion of the flow path **226** between the inner radial surface **228** and the outer radial surface **230**. For example, the second variable inlet guide vane **404** can span from the outer radial surface **230** and occupy from approximately 50% to approximately 10% of the radial height of the flow path

226. In FIG. 4, the second leading edge **416** is positioned downstream of the first trailing edge **408**. [0057] In FIG. 4, the first shaft **414** and the second shaft **424** are coupled to one or more actuators, which adjust a pitch of the first variable inlet guide vane **402** along a first trajectory **426** and the second variable inlet guide vane **404** along a second trajectory **428**. As a result, adjustments to the pitches of the variable inlet guide vanes **402**, **404** can adjust the tip incidence encountered by downstream rotor blades (e.g., the LP compressor rotor blades **148**, the HP compressor rotor blades **152** of FIG. 1) and, thus, control (e.g., reduce) nonsynchronous vibration encountered by the rotor blades. In this example, the first shaft **414** defines a first rotational axis **430**, and the second shaft **424** defines a second rotational axis **432** downstream of the first rotational axis **430**.

[0058] In some examples, the first variable inlet guide vane **402** and the second variable inlet guide vane **404** are actuated together in opposite directions or in different proportions to adjust the tip incidence and, thus, control (e.g., reduce) nonsynchronous vibration encountered by the compressor rotor blades. In some other examples, the first variable inlet guide vane **402** and the second variable inlet guide vane **404** are independently actuated. Example actuation mechanisms associated with the stub-tandem variable inlet guide vanes **400** of FIG. 4 are discussed further in association with FIGS. 5, 6, 7, 8A-8D, 9, 10, 11, 12, 13, and/or 14.

[0059] In some examples, the second leading edge **416** is positioned as close to the first trailing edge **408** in the axial direction A as manufacturing tolerances permit without causing the vanes **402**, **404** to rub when rotating. In some other examples, the second leading edge **416** contacts and/or overlaps with the first variable inlet guide vane **402** in the axial direction A.

[0060] FIG. 5 illustrates a first actuation system **500** to adjust a pitch of the variable inlet guide vanes **200**, **300**, **400** of FIGS. 2, 3, and/or 4. The first actuation system **500** includes an actuator **502**, a first ring **504** (e.g., an outer ring), a second ring **506** (e.g., an inner ring), a gear **508**, a first lever arm **510**, and a second lever arm **512**. The first lever arm **510** couples a first shaft **514** (e.g., the first shaft **214** of FIG. 2, the first shaft **318** of FIG. 3, the first shaft **414** of FIG. 4) of a first variable inlet guide vane (e.g., the first variable inlet guide vane **202** of FIG. 2, the first variable inlet guide vane **302** of FIG. 3, the first variable inlet guide vane **402** of FIG. 4) to the first ring **504**. The second lever arm **512** couples a second shaft **516** (e.g., the second shaft **224** of FIG. 2, the second shaft **332** of FIG. 3, the second shaft **424** of FIG. 4) of a second variable inlet guide vane (e.g., a stub, the second variable inlet guide vane **204** of FIG. 2, the second variable inlet guide vane **304** of FIG. 3, the second variable inlet guide vane **404** of FIG. 4) to the second ring **506**. In this example, the actuation system **500** is positioned radially outward of the flow path **226** of FIGS. 2-4. For example, the actuation system **500** can be coupled to the outer casing **110** of FIG. 1, as discussed further in association with FIG. 6.

[0061] In FIG. 5, the actuator **502** is operatively coupled to the first ring **504** or the second ring **506**. In some examples, the actuator **502** is implemented by a hydraulic pump or a servo motor that is operatively coupled to the first ring **504** or the second ring **506** via one or more bell cranks and/or crank and slider linkages. In FIG. 5, when the actuator **502** is coupled to the first ring **504**, the actuator **502** can cause the first ring **504** to rotate in a first direction (e.g., clockwise), which causes the gear **508** to rotate in the first direction between the first ring **504** and the second ring **506**. Further, the rotation of the gear **508** can cause the second ring **506** to rotate in a second direction (e.g., counterclockwise) opposite the first direction. Alternatively, when the actuator **502** is coupled to the second ring **506**, the actuator **502** can cause the second ring **506** to rotate in the first direction or the second direction, which also causes the gear **508** and, in turn, the first ring **504** to rotate in the opposite direction. The rotation of the rings **504**, **506** pivots the lever arms **510**, **512**, which, in turn, rotates the shafts **514**, **516** to adjust a pitch of the vanes connected to the shafts **514**, **516**. As such, the actuation system **500** of FIG. 5 rotates the shafts **514**, **516** and their associated vanes in opposite directions to control the tip incidence encountered by downstream rotor blades (e.g., the LP compressor rotor blades **148**, the HP compressor rotor blades **152** of FIG. 1) and, thus, control (e.g., reduce) nonsynchronous vibration encountered by the rotor blades.

[0062] Although the shafts **514**, **516** are shown in different axial positions in the illustrated example of FIG. 5, it should be understood that the shafts **514**, **516** can be positioned in a same axial position when the actuation system **500** is utilized in conjunction with the variable inlet guide vanes **200** of FIG. 2. In such examples, the shaft **514**, **516** that is positioned concentrically within the other shaft **516**, **514** extends further radially outward to provide a surface to which the lever arm **510**, **512** can attach.

[0063] FIG. 6 illustrates another view of the example actuation system **500** of FIG. 5. Specifically, FIG. 6 is a schematic illustration of an example axial cross-section of an engine (e.g., the engine **100** of FIG. 1) that includes the example actuation system **500** of FIG. 5. The actuation system **500** includes first linkages **602** to couple the first ring **504** to the outer casing **110**, second linkages **604** to couple the second ring **506** to the outer casing **110**, and third linkages **606** to couple the gears **508** to the outer casing **110**. The first ring **504** is aligned with the second ring **506** in the axial direction A and positioned radially outward of the second ring **506**. That is, the first ring **504** and the second ring **506** are concentric circles. In the illustrated example of FIG. 6, the actuator **502**, the lever arms **510**, **512**, and the shafts **514**, **516** are removed to show the linkages **602**, **604**, **606**, the rings **504**, **506**, and the gear **508**.

[0064] In FIG. 6, the first linkages **602**, the second linkages **604**, and the third linkages **606** enable the outer casing **110** to support a load of the rings **504**, **506** and the gears **508**. The first linkages **602** are circumferentially spaced apart to support the first ring **504**. Similarly, the second linkages **604** are circumferentially spaced apart to support the second ring **506**. The first linkages **602** include an outer radial end **608** and an inner radial end **610**. In some examples, the outer radial end **608** is pivotably coupled to the first ring **504**. In some examples, the inner radial end **610** is pivotably coupled to the outer casing **110**. In some other examples, the inner radial end **610** is fixedly coupled to the outer casing **110**. In such examples, the first linkages **602** include a link (not shown, extending from the outer radial end **608** into the page) to couple the outer radial end **608** to the first ring **504**. In such examples, a portion of the first linkages **602** from the inner radial end **610** to the outer radial end **608** is fixed in position and the link extending from the outer radial end **608** to the first ring **504** is pivotable about the outer radial end **608** to provide support while enabling the first ring **504** to rotate. For example, the link of the first linkages **602** can be coupled to the outer radial end **608** via a pin that enables the link to pivot about the outer radial end **608**. Further, an opposite end of the link can be coupled to the first ring **504** via another pin that enables an end of the link that couples to the first ring **504** for support to also pivot about the pin as the first ring **504** rotates. As such, the first linkages **602** enable the first ring **504** to rotate relative to the outer casing **110** to cause the shafts **514** (FIG. 5) and, in turn, the variable inlet guide vanes connected to the shafts **514** to rotate.

[0065] The second linkages **604** can be similar to the first linkages **602**. That is, the second linkages **604** includes an outer radial end **612** and an inner radial end **614**. In some examples, the outer radial end **612** is pivotably coupled to the second ring **506**. In some examples, the inner radial end **614** is pivotably coupled to the outer casing **110**. In some other examples, the inner radial end **614** is fixedly coupled to the outer casing **110**. In such examples, the second linkages **604** include a link (not shown, extending from the outer radial end **612** into the page) to couple the outer radial end **612** to the second ring **506**. In such examples, a portion of the second linkages **604** from the inner radial end **614** to the outer radial end **612** is fixed in position and the link extending from the outer radial end **612** to the second ring **506** is pivotable about the outer radial end **612** to provide support while enabling the second ring **506** to rotate. For example, the link of the second linkages **604** can be coupled to the outer radial end **612** via a pin that enables the link to pivot about the outer radial end **612**. Further, an opposite end of the link can be coupled to the second ring **506** via another pin that enables an end of the link that couples to the second ring **506** for support to also pivot about the pin as the second ring **506** rotates. As such, the second linkages **604** enable the second ring **506** to rotate relative to the outer casing **110** to cause the shafts **516** (FIG. 5) and, in turn, the variable

inlet guide vanes connected to the shafts **516** to rotate.

[0066] In FIG. **6**, the third linkages **606** support the gears **508**. For example, the third linkages **606** can be fixedly coupled to the outer casing **110** such that the third linkages **606** are immovable relative to the outer casing **110**. In such examples, the third linkages **606** are coupled to and/or include a rod or pin that extends through the gears **508** at a rotational axis of the gears. As such, the third linkages **606** support the gears **508** and maintain the position of the gears relative to the outer casing **110**. Accordingly, a rotation of the first ring **504** or the second ring **506** can cause the gears **508** to rotate in place to push the other ring **504**, **506** in the opposite direction.

[0067] In FIG. **6**, the actuation system **500** also includes fourth linkages **616** to couple the first ring **504** to the second ring **506**. For example, the fourth linkages **616** can include a first end **618** pivotably coupled to the first ring **504** and a second end **620** pivotably coupled to the second ring **506**. In some examples, the fourth linkages **616** limit a rotational displacement between the rings **504**, **506**. In some examples, the fourth linkages **616** relay support of the first ring **504** to the second ring **506** or support of the second ring **506** to the first ring **504** to enable the first linkages **602** or the second linkages **604** to be removed.

[0068] FIG. **7** illustrates another example actuation system **700** to adjust a pitch of a first variable inlet guide vane **702** (e.g., a full-span vane, a first variable pitch airfoil) and a second variable inlet guide vane **704** (e.g., a stub, a part-span vane, a second variable pitch airfoil). The first and second variable inlet guide vanes **702**, **704** of FIG. **7** are similar to the variable inlet guide vanes **300** of FIG. **3**. In the illustrated example of FIG. **7**, an axial position of a rotational axis **706** of the second variable inlet guide vane **704** overlaps with an axial position of the first variable inlet guide vane **702**. However, it should be understood that the actuation system **700** can be utilized in association with the variable inlet guide vanes **300** of FIG. **3** in which an axial position of the second rotational axis **340** (FIG. **3**) of the second variable inlet guide vane **304** (FIG. **3**) is positioned downstream of and does not overlap with the axial position of the first variable inlet guide vane **302** (FIG. **3**). Additionally, the example actuation system **700** of FIG. **7** may be utilized with the variable inlet guide vanes **400** of FIG. **4**.

[0069] In FIG. **7**, the actuation system **700** also includes the outer casing **110** and a bushing **708** between an outer radial edge **713** of the second variable inlet guide vane **704**. The outer casing **110** helps facilitate movement (e.g., rotation, an adjustment to an angular displacement) of the second variable inlet guide vane **704** relative to the outer casing **110**. More particularly, the second variable inlet guide vane **704** includes a spindle **711** that is positioned in and rotatably coupled to the outer casing **110**. The bushing **708** separates the spindle **711** and the outer radial edge **713** of the second variable inlet guide vane **704** from the outer casing **110**. The actuation system **700** also includes a lever arm **712** coupled to a shaft **714** (e.g., the first shaft **318** of FIG. **3**) of the first variable inlet guide vane **702**. The actuation system **700** further includes an actuator **716** operatively coupled to the lever arm **712** to control rotation of the shaft **714**. In some examples, the actuator **716** is operatively coupled to the lever arm **712** via a ring (e.g., the ring **504** (FIG. **5**)) and/or other linkages (e.g., one or more bell cranks, one or more crank and slider linkages, etc.). The actuator **716** can be implemented by a hydraulic pump or a servo motor.

[0070] In the illustrated example of FIG. **7**, an upper trailing edge **718** (e.g., the second trailing edge **310** (FIG. **3**) of the first variable inlet guide vane **702** and a leading edge **720** (e.g., the second leading edge **324** (FIG. **3**)) of the second variable inlet guide vane **704** include an interface **721**, as discussed further in association with FIGS. **8A-8D**. That is, the upper trailing edge **718** or a surface of the first variable inlet guide vane **702** is in contact with and/or overlaps in the axial direction **A** the leading edge **720** of the second variable inlet guide vane **704**. As a result, a rotation of the shaft **714** causes the upper trailing edge **718** or a surface of the first variable inlet guide vane **702** to push against the second variable inlet guide vane **704**, which produces a moment about the rotational axis **706** to turn the second variable inlet guide vane **704**. Thus, when the actuator **716** causes the first variable inlet guide vane **702** to rotate in a first direction (e.g., clockwise), the first variable

inlet guide vane **702** causes the second variable inlet guide vane **704** to rotate in a second direction opposite the first direction (e.g., counterclockwise).

[0071] FIGS. **8A-8B** illustrate a first example interface **800** (e.g., a first example implementation of the interface **721** of FIG. **7**) between the variable inlet guide vanes **702**, **704** of FIG. **7** that causes rotation of the first variable inlet guide vane **702** to produce a rotation of the second variable inlet guide vane **704**. FIG. **8A** illustrates a side view of the interface **800** between the first variable inlet guide vane **702** and the second variable inlet guide vane **704**. In FIG. **8A**, the upper trailing edge **718** of the first variable inlet guide vane **702** includes first protrusions **802** (e.g., tabs, projections, ridges, etc.) and first slots **804** defined between the first protrusions **802**. The leading edge **720** of the second variable inlet guide vane **704** includes second protrusions **806** and second slots **808** defined between the second protrusions **806**. More particularly, the protrusions **802**, **806** are spaced apart in the radial direction **R** to define the slots **804**, **808** between the protrusions **802**, **806**. The first protrusions **802** of the first variable inlet guide vane **702** are positioned in the second slots **808** of the second variable inlet guide vane **704**. Similarly, the second protrusions **806** of the second variable inlet guide vane **704** are positioned in the first slots **804** of the first variable inlet guide vane **702**. The protrusions **802**, **806** include orifices in which a pin is positioned to relay a rotation of the first variable inlet guide vane **702** to a rotation of the second variable inlet guide vane **704**.

[0072] FIG. **8B** illustrates a schematic top-down (e.g., radially inward) view of the interface **800** of FIG. **8A**. The view of FIG. **8B** is an idealized schematic view provided for purposes of illustration. In practice, portions of the protrusions **802**, **806** that overlap in the axial direction **A** are stacked in the radial direction **R** such that the second protrusion **806** is positioned under the first protrusion **802** in a top-down view. As mentioned above, the first protrusion **802** includes a first orifice **810** and the second protrusion **806** includes a second orifice **812**, and the interface **800** includes a pin **814** that is positioned in and extends through the orifices **810**, **812**. The protrusions **802**, **806** are stacked to align the orifices **810**, **812** and enable the pin **814** to pass through the orifices **810**, **812** in the radial direction **R**. That is, the orifices **810**, **812** define a slot in which the pin **814** is positioned. As a result, the pin **814** can relay movement of the upper trailing edge **718** to movement of the leading edge **720** that causes the second variable inlet guide vane **704** to rotate about the rotational axis **706** (FIG. **7**). More particularly, as the actuator **716** (FIG. **7**) rotates the first variable inlet guide vane **702** and causes the upper trailing edge **718** to move in a first direction (e.g., into the page in the view of FIG. **7**), the pin **814** moves with the first protrusions **802** and applies a force to the second protrusions **806** that moves the leading edge **720** in the first direction. To enable the leading edge **720** to move in the first direction, the second variable inlet guide vane **704** rotates about the rotational axis **706** (FIG. **7**). Furthermore, the rotation causes a trailing edge (not shown) (e.g., the third trailing edge **326**) of the second variable inlet guide vane **704** to move in a second direction opposite the first direction (e.g., out of the page in the view of FIG. **7**). As such, the interface **800** causes a counterclockwise rotation of the first variable inlet guide vane **702** to produce a clockwise rotation of the second variable inlet guide vane **704**. In some examples, the orifices **810**, **812** provide space for some movement of the pin **814** in the axial direction **A** to increase a range of motion of the variable inlet guide vanes **702**, **704**.

[0073] FIGS. **8C-8D** illustrate a second example interface **850** (e.g., a second example implementation of the interface **721** of FIG. **7**) of the variable inlet guide vanes **702**, **704** of FIG. **7** that causes rotation of the first variable inlet guide vane **702** to produce a rotation of the second variable inlet guide vane **704**. FIG. **8C** illustrates a side or circumferential perspective of the interface **850** between the first variable inlet guide vane **702** and the second variable inlet guide vane **704**. FIG. **8D** illustrates a top-down (e.g., radially inward) view of the interface **850** of FIG. **8A**. In FIG. **8D**, the upper trailing edge **718** of the first variable inlet guide vane **702** includes an indentation **852** that defines a slot **854**. The leading edge **720** of the second variable inlet guide vane **704** includes a protrusion **856** that is positioned in the slot **854**. As such, when the first variable inlet guide vane **702** rotates, a surface of the indentation **852** contacts the protrusion **856** to

apply a force that pushes the leading edge **720** of the second variable inlet guide vane **704** in a direction in which the upper trailing edge **718** is moving. As a result, the movement of the leading edge **720** causes the second variable inlet guide vane **704** to rotate about the rotational axis **706** (FIG. 7) in a direction opposite of the rotational direction of the first variable inlet guide vane **702**. [0074] Although FIG. 4 mentions that the second leading edge **416** (FIG. 4) is positioned aft of the first trailing edge **408** (FIG. 4), in some examples, the edges **408**, **416** refer to ends of a side surface (e.g., a pressure surface, a suction surface). In such examples, the interface **850** can be utilized in conjunction with the third variable inlet guide vanes **400** of FIG. 4.

[0075] FIG. 9 illustrates another example actuation system **900** to adjust a pitch of a first variable inlet guide vanes **902** (e.g., the first variable inlet guide vane **302** of FIG. 3, the first variable inlet guide vane **402** of FIG. 4) and a second variable inlet guide vane **904** (e.g., the second variable inlet guide vane **304** of FIG. 3, the second variable inlet guide vanes **404** of FIG. 4). In FIG. 9, the actuation system **900** includes an actuator **905**, a pin **906**, a ring **908**, a first lever arm **910**, and a second lever arm **912**. In FIG. 9, the first lever arm **910** is coupled to a first spindle **914** (e.g., a first shaft) of the first variable inlet guide vane **902**. Similarly, the second lever arm **912** is coupled to a second spindle **916** (e.g., a second shaft) of the second variable inlet guide vane **904**. In some examples, the actuation system **900** is positioned radially outward of the outer casing **110**.

[0076] In FIG. 9, the actuator **905** controls a rotational position of the ring **908**. That is, to adjust an angular displacement of the first and second variable inlet guide vanes **902**, **904**, the actuator **905** can rotate the ring **908**. For example, the actuator **905** can be implemented by a hydraulic pump or a servo motor that is operatively coupled to the ring **908** via one or more bell cranks and/or crank and slider linkages (not shown) to enable the actuator **905** to control the rotational position of the ring **908**. When the actuator **905** rotates the ring **908**, the ring **908** moves the pin **906**. The movement of the pin **906** is linear (e.g., into and/or out of the page) in the view of FIG. 9. In reality, the pin **906** moves in the circumferential direction C within the engine **100** of FIG. 1. More particularly, when implemented in the engine **100** of FIG. 1, the movement causes the pin **906** to revolve around the axial centerline axis **108** of FIG. 1. To that end, when implemented in the engine **100** of FIG. 1, the ring **908** is positioned in and/or around the outer casing **110** and concentrically around the flow path **226** (FIGS. 2-4) of the second portion **144** of the air **136**. Further, the actuation system **900** includes a plurality of the pin **906** coupled to the ring **908** and circumferentially spaced apart to control the angular displacement of a plurality of the first and second variable inlet guide vanes **902**, **904** at the inlet **112** (FIG. 1).

[0077] In FIG. 9, the lever arms **910**, **912** are pivotably coupled to the pin **906**. That is, the first lever arm **910** includes (i) a first end **918** (e.g., a trailing end) pivotably coupled to the pin **906** and (ii) a second end **920** (e.g., a leading end) fixedly coupled to the first spindle **914**. Similarly, the second lever arm **912** includes (i) a first end **922** (e.g., a leading end) pivotably coupled to the pin **906** and (ii) a second end **924** (e.g., a trailing end) fixedly coupled to the second spindle **916**. As a result, as the pin **906** moves, the lever arms **910**, **912** rotate the spindles **914**, **916** and, thus, the first and second variable inlet guide vanes **902**, **904**. The actuation system **900** of FIG. 9 causes the first and second variable inlet guide vanes **902**, **904** to rotate in opposite directions. For example, when the ring **908** moves the pin **906** out of the page in the view of FIG. 9, the first lever arm **910** causes the first variable inlet guide vane **902** to rotate clockwise, and the second lever arm **912** causes the second variable inlet guide vane **904** to rotate counterclockwise.

[0078] FIG. 10 illustrates an example cross-section of a portion of the actuation system **900** of FIG. 9 taken along plane A-A of FIG. 9. Specifically, FIG. 10 illustrates a movement direction **1002** of the pin **906**. In some examples, the movement direction **1002** corresponds to the circumferential direction C of FIG. 1 when the actuation system **900** is implemented in the engine **100** of FIG. 1. In FIG. 10, as the pin **906** moves in the movement direction **1002**, the lever arms **910**, **912** rotate the spindles **914**, **916** to adjust a pitch of the first and second variable inlet guide vanes **902**, **904**.

[0079] FIG. 11 illustrates another example actuation system **1100** to adjust a pitch of a first variable

inlet guide vanes **1102** (e.g., the first variable inlet guide vane **302** of FIG. 3, the first variable inlet guide vane **402** of FIG. 4) and a second variable inlet guide vane **1104** (e.g., the second variable inlet guide vane **304** of FIG. 3, the second variable inlet guide vanes **404** of FIG. 4). In FIG. 11, the actuation system **1100** includes an actuator **1105**, a first pin **1106**, a ring **1108**, a mounting **1110** (e.g., a frame, a holder, a linkage, etc.), a second pin **1112**, a third pin **1114**, a first lever arm **1116**, a second lever arm **1118**, and a third lever arm **1120**. In FIG. 11, the first pin **1106** is rotatably coupled to the outer casing **110**. The actuator **1105** controls a rotational position of the ring **1108**. That is, to adjust an angular displacement of the first and second variable inlet guide vanes **1102**, **1104**, the actuator **1105** can rotate the ring **1108** in the circumferential direction C defined by the engine **100** of FIG. 1. For example, the actuator **1105** can be implemented by a hydraulic pump or a servo motor that is operatively coupled to the ring **1108** via one or more bell cranks and/or crank and slider linkages to enable the actuator **1105** to control the rotational position of the ring **1108**. In FIG. 11, the first lever arm **1116** converts a rotation of the ring **1108** to a rotation of the first pin **1106**.

[0080] In FIG. 11, the mounting **1110** is fixedly coupled to the first pin **1106**. As a result, the mounting **1110** rotates with the first pin **1106**. The second pin **1112** and the third pin **1114** are rotatably coupled to the mounting **1110**. Additionally, the second pin **1112** is rotatably coupled to the second lever arm **1118**, and the third pin **1114** is rotatably coupled to the third lever arm **1120**. The second lever arm **1118** is also fixedly coupled to the first variable inlet guide vane **1102** (e.g., a shaft or spindle of the first variable inlet guide vane **1102**). Similarly, the third lever arm **1120** is fixedly coupled to the second variable inlet guide vane **1104** (e.g., a shaft or spindle of the second variable inlet guide vane **1104**).

[0081] In FIG. 11, when the actuator **1105** rotates the ring **1108** and, in turn, the first pin **1106** and the mounting **1110**, the mounting **1110** revolves the second pin **1112** and the third pin **1114** around the first pin **1106**. As a result, the second pin **1112** and the third pin **1114** cause the second lever arm **1118** and the third lever arm **1120** to pivot, which rotates the first variable inlet guide vane **1102** and the second variable inlet guide vane **1104**. The actuation system **1100** of FIG. 11 causes the first and second variable inlet guide vanes **1102**, **1104** to rotate in a same direction, as discussed further in association with FIG. 12.

[0082] FIG. 12 illustrates an example cross-section of a portion of the actuation system **1100** of FIG. 11 taken along plane B-B of FIG. 11. Specifically, FIG. 12 illustrates a movement direction **1202** of the first pin **1106** and the mounting **1110**. In FIG. 12, as the first pin **1106** rotates (e.g., in the clockwise direction in the view of FIG. 12), the mounting **1110** rotates about the first pin **1106**. As a result, the mounting **1110** moves the second pin **1112** to cause a leading end **1204** of the second lever arm **1118** to be positioned above a trailing end **1206** of the second lever arm **1118**. As a result, the second lever arm **1118** turns the first variable inlet guide vane **1102** in a clockwise direction. Similarly, the mounting **1110** moves the third pin **1114** to cause a leading end **1208** of the third lever arm **1120** to be positioned above a trailing end **1210** of the third lever arm **1120**. As a result, the third lever arm **1120** turns the second variable inlet guide vane **1104** in a clockwise direction. Thus, in the actuation system **1100**, the direction in which the first variable inlet guide vane **1102** rotates matches the direction in which the second variable inlet guide vane **1104** rotates. Additionally, the direction in which the first and second variable inlet guide vanes **1102**, **1104** rotate matches the movement direction of the first pin **1106** and the mounting **1110**.

[0083] In some examples, the actuation system **1100** causes the positions of the first and second variable inlet guide vanes **1102**, **1104** to be adjusted by a same angular displacement (e.g., the same magnitude (e.g., degrees)) ($^{\circ}$ in the clockwise direction). In some other examples, the actuation system **1100** causes the positions of the first and second variable inlet guide vanes **1102**, **1104** to be adjusted by different angular displacements (e.g., different magnitudes (e.g., degrees)) ($^{\circ}$ in the clockwise direction). For example, a length(s) of the second lever arm **1118** and/or the third lever arm **1120** can be adjusted to configure the magnitudes of angular displacement that the first and

second variable inlet guide vanes **1102**, **1104** encounter as a result of a unit of rotation of the first pin **1106** and the mounting **1110**. Further, a size and/or shape of the mounting **1110** can be adjusted based on the length(s) of the second and third lever arms **1118**, **1120** and the desired conversion between the rotation of the first pin **1106** and the angular displacement(s) of the first and second variable inlet guide vanes **1102**, **1104**.

[0084] FIG. **13** illustrates another example actuation system **1300** to adjust a pitch of a first variable inlet guide vanes **1302** (e.g., the first variable inlet guide vane **202** of FIG. **2**, the first variable inlet guide vane **302** of FIG. **3**, the first variable inlet guide vane **402** of FIG. **4**) and a second variable inlet guide vane **1304** (e.g., the second variable inlet guide vane **204** of FIG. **2**, the second variable inlet guide vane **304** of FIG. **3**, the second variable inlet guide vanes **404** of FIG. **4**). In FIG. **13**, the actuation system **1300** includes an actuator **1306**, a ring **1308**, a first lever arm **1310**, and a second lever arm **1312**. In FIG. **13**, the actuator **1306** controls a rotational position of the ring **1308**. That is, to adjust an angular displacement of the first and second variable inlet guide vanes **1302**, **1304**, the actuator **1306** can rotate the ring **1308**. For example, the actuator **1306** can be implemented by a hydraulic pump or a servo motor that is operatively coupled to the ring **1308** via one or more bell cranks and/or crank and slider linkages to enable the actuator **1306** to control the rotational position of the ring **1308**.

[0085] In FIG. **13**, the first lever arm **1310** and the second lever arm **1312** are pivotably coupled to the ring **1308**. In some examples, the first lever arm **1310** is pivotably coupled to the ring **1308** at an outer radial surface **1314** of the ring **1308**. In some examples, the second lever arm **1312** is pivotably coupled to the ring **1308** at an inner radial surface **1316** of the ring **1308**. Further the first lever arm **1310** is fixedly coupled to the first variable inlet guide vane **1302** (e.g., a shaft or spindle of the first variable inlet guide vane **1302**), and the second lever arm **1312** is fixedly coupled to the second variable inlet guide vane **1304** (e.g., a shaft or spindle of the second variable inlet guide vane **1304**).

[0086] In FIG. **13**, when the actuator **1306** rotates the ring **1308** (e.g., in the circumferential direction C of FIG. **1**), first ends **1318**, **1320** of the lever arms **1310**, **1312** (e.g., ends of the lever arms **1310**, **1312** pivotably coupled to the ring **1308**) move with the ring **1308**. As a result, second ends **1322**, **1324** of the lever arms **1310**, **1312** (e.g., ends of the lever arms **1310**, **1312** fixedly coupled to the variable inlet guide vanes **1302**, **1304**) rotate, which rotates the first variable inlet guide vane **1302** and the second variable inlet guide vane **1304**. In FIG. **13**, the actuation system **1300** causes the first variable inlet guide vane **1302** and the second variable inlet guide vane **1304** to rotate in a same direction. In some examples, the actuation system **1300** causes angular displacements of the first variable inlet guide vane **1302** and the second variable inlet guide vane **1304** to change at a same rate when the ring **1308** rotates. In some other examples, the actuation system **1300** causes angular displacements of the first variable inlet guide vane **1302** and the second variable inlet guide vane **1304** to change at different rates. For example, a length(s) of the lever arms **1310**, **1312** can be shortened to increase the change in angular displacement of the first variable inlet guide vane **1302** and the second variable inlet guide vane **1304** when the ring **1308** rotates.

[0087] Although shafts of the variable inlet guide vanes **1302**, **1304** are shown in different axial positions in the illustrated example of FIG. **13**, it should be understood that the shafts can be positioned in a same axial position when the actuation system **1300** is utilized in conjunction with the variable inlet guide vanes **200** of FIG. **2**. In such examples, the shaft that is positioned concentrically within the other shaft extends further radially outward to provide a surface to which the lever arms **1310**, **1312** can attach.

[0088] FIG. **14** illustrates another example actuation system **1400** to adjust a pitch of a first variable inlet guide vane **1402** (e.g., the first variable inlet guide vane **202** of FIG. **2**, the first variable inlet guide vane **302** of FIG. **3**, the first variable inlet guide vane **402** of FIG. **4**) and a second variable inlet guide vane **1404** (e.g., the second variable inlet guide vane **204** of FIG. **2**, the

second variable inlet guide vane **304** of FIG. **3**, the second variable inlet guide vanes **404** of FIG. **4**). In FIG. **14**, the actuation system **1400** includes an actuator **1406**, a ring **1407**, a first lever arm **1412**, and a second lever arm **1414**. The ring **1407** includes an outer radial portion **1408** and an inner radial portion **1410**. In FIG. **14**, the actuator **1406** controls a rotational position of the ring **1407**. That is, to adjust an angular displacement of the first variable inlet guide vane **1402** and the second variable inlet guide vane **1404**, the actuator **1406** can rotate the ring **1407**. For example, the actuator **1306** can be implemented by one or more hydraulic pumps or a servo motors that are operatively coupled to the ring **1407** via one or more bell cranks and/or crank and slider linkages to enable the actuator **1406** to control the rotational position of the ring **1407**.

[0089] In FIG. **14**, a first end **1416** of the first lever arm **1412** is pivotably coupled to the outer radial portion **1408** of the ring **1407**. A second end **1418** of the first lever arm **1412** is fixedly coupled to the first variable inlet guide vane **1402** (e.g., a shaft or spindle of the first variable inlet guide vane **1402**). Similarly, a first end **1420** of the second lever arm **1414** is pivotably coupled to the inner radial portion **1410** of the ring **1407**, and a second end **1422** of the second lever arm **1414** is fixedly coupled to the second variable inlet guide vane **1404** (e.g., a shaft or spindle of the second variable inlet guide vane **1404**). In some examples, the first end **1416** of the first lever arm **1412** is coupled to an outer radial surface **1424** of the ring **1407**. In some examples, the first end **1420** of the second lever arm **1414** is coupled to an inner radial surface **1426** of the ring **1407**.

[0090] In FIG. **14**, when the actuator **1406** rotates the ring **1407** (e.g., in the circumferential direction C of FIG. **1**), the first end **1416** of the first lever arm **1412** moves with the outer radial portion **1408** of the ring **1407**. As a result, the second end **1418** of the first lever arm **1412** rotates, which rotates the first variable inlet guide vane **1402**. Similarly, when the actuator **1406** rotates the ring **1407** (e.g., in the circumferential direction C of FIG. **1**), the first end **1420** of the second lever arm **1414** moves with the inner radial portion **1410**, and the second end **1422** of the second lever arm **1414** rotates, which rotates the second variable inlet guide vane **1404**.

[0091] In some examples, the variable inlet guide vanes **1402**, **1404** rotate in the same direction. In FIG. **14**, the outer radial portion **1408** of the ring **1407** includes a different geometry than the inner radial portion **1410** to enable a desired angular displacement relationship. More particularly, the inner radial portion **1410** is offset from the outer radial portion **1408** in an axial direction A, which enables the second lever arm **1414** to be shorter than the first lever arm **1412**. As a result, a same magnitude of rotation by the outer radial portion **1408** and the inner radial portion **1410** can cause the second variable inlet guide vane **1404** to rotate more than the first variable inlet guide vane **1402**.

[0092] Although shafts of the variable inlet guide vanes **1402**, **1404** are shown in different axial positions in the illustrated example of FIG. **14**, it should be understood that the shafts can be positioned in a same axial position when the actuation system **1400** is utilized in conjunction with the variable inlet guide vanes **200** of FIG. **2**. In such examples, the shaft that is positioned concentrically within the other shaft extends further radially outward to provide a surface to which the lever arms **1412**, **1414** can attach.

[0093] From the foregoing, it will be appreciated that example systems, apparatus, articles of manufacture, and methods have been disclosed that reduce non-synchronous vibration that can be encountered in a compressor. As a result, examples herein suppress formation of rotating instabilities associated with non-synchronous vibration and improve engine performance.

[0094] Example gas turbine engines including stub-tandem variable inlet guide vanes are disclosed. Further aspects are provided by the subject matter of the following clauses:

[0095] A gas turbine engine comprising a compressor, a variable inlet guide vane upstream of the compressor, and a part-span vane positioned between at least a portion of the variable inlet guide vane and the compressor in an axial direction defined by the gas turbine engine.

[0096] The gas turbine engine of the preceding clause, wherein the part-span vane includes a first portion and a second portion downstream of the first portion, wherein the first portion overlaps

with the variable inlet guide vane in the axial direction, and wherein the second portion is aft of the variable inlet guide vane in the axial direction.

[0097] The gas turbine engine of any preceding clause, wherein the variable inlet guide vane includes an upstream portion and a downstream portion, wherein the upstream portion spans a greater distance than the downstream portion in a radial direction defined by the gas turbine engine.

[0098] The gas turbine engine of any preceding clause, wherein the part-span vane is a variable part-span vane.

[0099] The gas turbine engine of any preceding clause, wherein the variable inlet guide vane includes a first shaft coupled to at least one actuator, wherein the variable part-span vane includes a second shaft coupled to the at least one actuator, and wherein the second shaft is positioned aft of the first shaft.

[0100] The gas turbine engine of any preceding clause, wherein the variable part-span vane rotates in a first direction when the variable inlet guide vane rotates in a second direction opposite the first direction.

[0101] The gas turbine engine of any preceding clause, wherein the variable part-span vane rotates in a first direction when the variable inlet guide vane rotates in the first direction.

[0102] The gas turbine engine of any preceding clause, wherein the variable inlet guide vane and the part-span vane are positioned in a flow path defined between an inner radial surface and an outer radial surface, wherein the variable inlet guide vane includes a first inner radial edge at the inner radial surface and a second outer radial edge at the outer radial surface.

[0103] The gas turbine engine of any preceding clause, wherein the variable inlet guide vane includes a first outer radial edge positioned between the first inner radial edge and the second outer radial edge in a radial direction defined by the gas turbine engine.

[0104] The gas turbine engine of any preceding clause, wherein the first outer radial edge is downstream of the second outer radial edge.

[0105] The gas turbine engine of any preceding clause, wherein the part-span vane overlaps the third radial end in the axial direction.

[0106] An apparatus comprising a fan, a compressor downstream of the fan, and stub-tandem variable inlet guide vanes positioned between the fan and the compressor, the stub-tandem variable inlet guide vanes including a first variable inlet guide vane upstream of the compressor, and a second variable inlet guide vane upstream of the compressor and downstream of at least a portion of the first variable inlet guide vane.

[0107] The apparatus of any preceding clause, wherein the second variable inlet guide vane is a part-span airfoil.

[0108] The apparatus of any preceding clause, wherein the second variable inlet guide vane is cantilevered.

[0109] The apparatus of any preceding clause, wherein the first variable inlet guide vane has a first pitch, and wherein the second variable inlet guide vane has a second pitch different from the first pitch.

[0110] The apparatus of any preceding clause, wherein an aft portion of the first variable inlet guide vane contacts a forward portion of the second variable inlet guide vane.

[0111] The apparatus of any preceding clause, further including an actuator operatively coupled to the first variable inlet guide vane, wherein the actuator is to cause the first variable inlet guide vane to rotate, and wherein the second variable inlet guide vane rotates in an opposite direction from the first variable inlet guide vane when the actuator causes the first variable inlet guide vane to rotate.

[0112] An apparatus comprising a first variable inlet guide vane including a first inner radial edge and a first outer radial edge, wherein a first distance from the first inner radial edge to the first outer radial edge defines a height of a flow path, and a second variable inlet guide vane downstream of the first variable inlet guide vane in the flow path, wherein the second variable inlet guide vane includes a second inner radial edge and a second outer radial edge, and wherein a second distance

from the second inner radial edge to the second outer radial edge is less than the first distance.

[0113] The apparatus of any preceding clause, wherein the first variable inlet guide vane includes a first leading edge and a first trailing edge, and wherein the second variable inlet guide vane includes a second leading edge positioned downstream of the first leading edge and upstream of the first trailing edge in the flow path.

[0114] The apparatus of any preceding clause, wherein the first variable inlet guide vane includes a third outer radial edge positioned between the first inner radial edge and the second outer radial edge in a radial direction that defines the height of the flow path.

[0115] An apparatus comprising, means for compressing, first means for variably producing aerodynamic forces upstream of the means for compressing, and second means for variably producing aerodynamic forces upstream of the means for compressing and at least partially downstream of the first means for variably producing aerodynamic forces.

[0116] The apparatus of any preceding clause, wherein the first means for variably producing aerodynamic forces spans a first distance of a radial length of a flow path, and wherein the second means for variably producing aerodynamic forces spans a second distance of a radial length of the flow path, the second distance different from the first distance.

[0117] The apparatus of any preceding clause, further including means for actuating the first means for variably producing aerodynamic forces and the second means for variably producing aerodynamic forces.

[0118] The apparatus of any preceding clause, wherein the means for actuating rotates the first means for variably producing aerodynamic forces and the second means for variably producing aerodynamic forces simultaneously in different directions.

[0119] The apparatus of any preceding clause, wherein the means for actuating rotates the first means for variably producing aerodynamic forces and the second means for variably producing aerodynamic forces simultaneously in a same direction.

[0120] The apparatus of any preceding clause, wherein the means for actuating rotates the first means for variably producing aerodynamic forces independent of the second means for variably producing aerodynamic forces.

[0121] The apparatus of any preceding clause, wherein the means for actuating rotates the first means for variably producing aerodynamic forces to a first angular displacement, and wherein the means for actuating rotates the second means for variably producing aerodynamic forces to a second angular displacement different from the first angular displacement.

[0122] An apparatus comprising, a first variable inlet guide vane to extend across a first radial portion of a flow path, a second variable inlet guide to extend across a second radial portion of a flow path, the second radial portion different from the first radial portion.

[0123] The apparatus of any preceding clause, further including an actuator to rotate the first variable inlet guide vane in a first direction, and wherein the actuator is to rotate the second variable inlet guide vane in a second direction opposite the first direction.

[0124] The apparatus of any preceding clause, further including an actuator to rotate the first variable inlet guide vane and the second variable inlet guide vane in a same direction at a same time.

[0125] The apparatus of any preceding clause, wherein the actuator rotates the first variable inlet guide to a different angular placement than the second variable inlet guide vane.

[0126] The apparatus of any preceding clause, further including one or more actuators to rotate the first variable inlet guide vane independent of the second variable inlet guide vane.

[0127] An apparatus comprising, a first variable inlet guide vane including a first portion and a second portion, the first portion having a shorter axial length than the second portion, and a second variable inlet guide vane downstream of and aligned with the second portion of the first variable inlet guide vane in a radial direction, the second variable inlet guide vane not aligned with the first portion of the first variable inlet guide vane in the radial direction.

[0128] An apparatus comprising, as first variable inlet guide spanning a radial length of a flow path, and a second variable inlet guide vane downstream of the first variable inlet guide vane and spanning only a portion of the radial length of the flow path.

[0129] The apparatus of any preceding clause, wherein the portion is an outer radial portion of the flow path.

[0130] An apparatus comprising, an actuator, a first ring, a second ring coupled to the actuator, wherein the first ring or the second ring is coupled to the actuator, a gear positioned between the first ring and the second ring to relay rotation of the first ring or the second ring by the actuator to the second ring or the first ring, a first variable inlet guide vane, a second variable inlet guide vane, a first lever arm including a first end pivotably coupled to the first ring and a second end fixedly coupled to the first variable inlet guide vane, a second lever arm including a third end pivotably coupled to the second ring and a second end fixedly coupled to the first ring.

[0131] The apparatus of any preceding clause, further including a casing, a first linkage including a first end coupled to the casing and a second end coupled to the first ring.

[0132] The apparatus of any preceding clause, further including a second linkage including a first end coupled to the casing and a second end coupled to the second ring.

[0133] The apparatus of any preceding clause, further including a third linkage including a first end coupled to the casing and a second end coupled to the gear between the first ring and the second ring.

[0134] The apparatus of any preceding clause, further including a fourth linkage including a first end coupled to the first ring and a second end coupled to the second ring.

[0135] An apparatus comprising, a first variable inlet guide vane, a second variable inlet guide aft of at least a portion of the first variable inlet guide vane, wherein a trailing edge of the first variable inlet guide vane overlaps with an axial position of the second variable inlet guide vane at an interface.

[0136] The apparatus of any preceding clause, further including an actuator to rotate the first variable inlet guide vane, and wherein the interface causes the second variable inlet guide vane to rotate in an opposite direction from the first variable inlet guide vane.

[0137] The apparatus of any preceding clause, wherein the first variable inlet guide vane and the second variable inlet guide vane include orifices through which a pin extends at the interface.

[0138] The apparatus of any preceding clause, wherein the first variable inlet guide vane includes an indentation, and wherein the second variable inlet guide vane includes a protrusion that extends into a slot defined by the indentation at the interface.

[0139] An apparatus comprising, a first variable inlet guide vane, a second variable inlet guide aft of at least a portion of the first variable inlet guide vane, an actuator, a ring coupled to the actuator, the actuator to rotate the ring, a pin fixedly coupled to the ring, a first lever arm pivotably coupled to the pin and fixedly coupled to the first variable inlet guide vane, and a second lever arm pivotably coupled to the pin and fixedly coupled to the second variable inlet guide vane.

[0140] The apparatus of any preceding clause, wherein a rotation of the ring in a first rotational direction causes the first inlet guide vane and the second variable inlet guide vane to rotate in different directions.

[0141] An apparatus comprising, a first variable inlet guide vane, a second variable inlet guide aft of at least a portion of the first variable inlet guide vane, an actuator, a ring coupled to the actuator, the actuator to rotate the ring, a first lever arm pivotably coupled to the ring, a first pin, the first lever arm to rotate the pin when the ring rotates, a mounting fixedly coupled to the first pin, the mounting to rotate with the first pin, a second pin coupled to the mounting, a third pin coupled to the mounting, a second lever arm to couple the first variable inlet guide vane to the second pin, and a third lever arm to couple the second variable inlet guide vane to the third pin.

[0142] The apparatus of any preceding clause, wherein a rotation of the ring in a first rotational direction causes the first inlet guide vane and the second variable inlet guide vane to rotate in a

same direction.

[0143] The apparatus of any preceding clause, wherein a rotation of the ring in a first rotational direction causes the first inlet guide vane to change angular displacement by a first degree or magnitude and causes the second variable inlet guide vane change angular displacement by a second degree or magnitude different from the first degree or magnitude.

[0144] The apparatus of any preceding clause, wherein a rotation of the ring in a first rotational direction causes the first inlet guide vane to change angular displacement by a same amount as the second variable inlet guide vane.

[0145] An apparatus comprising, a first variable inlet guide vane, a second variable inlet guide, an actuator, a ring, a first lever arm to couple the first variable inlet guide vane to the ring, and a second lever arm to couple the second variable inlet guide vane to the ring.

[0146] The apparatus of any preceding clause, wherein a rotation of the ring in a first rotational direction causes the first inlet guide vane and the second variable inlet guide vane to rotate in a same direction.

[0147] The apparatus of any preceding clause, wherein a rotation of the ring in a first rotational direction causes the first inlet guide vane to change angular displacement by a first degree or magnitude and causes the second variable inlet guide vane change angular displacement by a second degree or magnitude different from the first degree or magnitude.

[0148] The apparatus of any preceding clause, wherein a rotation of the ring in a first rotational direction causes the first inlet guide vane to change angular displacement by a same amount as the second variable inlet guide vane.

[0149] An apparatus comprising, a first variable inlet guide vane, a second variable inlet guide, at least one actuator, a first ring coupled to the at least one actuator, a second ring coupled to the at least one actuator, a first lever arm to couple the first variable inlet guide vane to the first ring, a second lever arm to couple the second variable inlet guide vane to the second ring.

[0150] The apparatus of any preceding clause, wherein the at least one actuator causes a first rotation of the first ring in a first direction, and wherein the at least one actuator causes a second rotation of the second ring in a second direction opposite the first direction, the second rotation to occur at a same time as the first rotation.

[0151] The apparatus of any preceding clause, wherein the at least one actuator causes a first rotation of the first ring in a first direction, and wherein the at least one actuator causes a second rotation of the second ring in the first direction, the second rotation to occur at a same time as the first rotation.

[0152] The apparatus of any preceding clause, wherein the at least one actuator causes the first inlet guide vane to change angular displacement by a first degree or magnitude and causes the second variable inlet guide vane change angular displacement by a second degree or magnitude different from the first degree or magnitude.

[0153] The following claims are hereby incorporated into this Detailed Description by this reference. Although certain example systems, apparatus, articles of manufacture, and methods have been disclosed herein, the scope of coverage of this patent is not limited thereto. On the contrary, this patent covers all systems, apparatus, articles of manufacture, and methods fairly falling within the scope of the claims of this patent.

Claims

1. A gas turbine engine comprising: a compressor; a variable inlet guide vane upstream of the compressor; and a part-span vane positioned between at least a portion of the variable inlet guide vane and the compressor in an axial direction defined by the gas turbine engine.
2. The gas turbine engine of claim 1, wherein the part-span vane includes a first portion and a second portion downstream of the first portion, wherein the first portion overlaps with the variable

inlet guide vane in the axial direction, and wherein the second portion is aft of the variable inlet guide vane in the axial direction.

3. The gas turbine engine of claim 1, wherein the variable inlet guide vane includes an upstream portion and a downstream portion, wherein the upstream portion spans a greater distance than the downstream portion in a radial direction defined by the gas turbine engine.

4. The gas turbine engine of claim 1, wherein the part-span vane is a variable part-span vane.

5. The gas turbine engine of claim 4, wherein the variable inlet guide vane includes a first shaft coupled to at least one actuator, wherein the variable part-span vane includes a second shaft coupled to the at least one actuator, and wherein the second shaft is positioned aft of the first shaft.

6. The gas turbine engine of claim 4, wherein the variable part-span vane rotates in a first direction when the variable inlet guide vane rotates in a second direction opposite the first direction.

7. The gas turbine engine of claim 4, wherein the variable part-span vane rotates in a first direction when the variable inlet guide vane rotates in the first direction.

8. The gas turbine engine of claim 1, wherein the variable inlet guide vane and the part-span vane are positioned in a flow path defined between an inner radial surface and an outer radial surface, wherein the variable inlet guide vane includes a first inner radial edge at the inner radial surface and a second outer radial edge at the outer radial surface.

9. The gas turbine engine of claim 8, wherein the variable inlet guide vane includes a first outer radial edge positioned between the first inner radial edge and the second outer radial edge in a radial direction defined by the gas turbine engine.

10. The gas turbine engine of claim 9, wherein the first outer radial edge is downstream of the second outer radial edge.

11. The gas turbine engine of claim 10, wherein the part-span vane overlaps the first outer radial edge in the axial direction.

12. An apparatus comprising: a fan; a compressor downstream of the fan; and stub-tandem variable inlet guide vanes positioned between the fan and the compressor, the stub-tandem variable inlet guide vanes including: a first variable inlet guide vane upstream of the compressor; and a second variable inlet guide vane upstream of the compressor and downstream of at least a portion of the first variable inlet guide vane.

13. The apparatus of claim 12, wherein the second variable pitch airfoil spans only a portion of a radial height of a flow path between the fan and the compressor.

14. The apparatus of claim 12, wherein the second variable inlet guide vane is cantilevered.

15. The apparatus of claim 12, wherein the first variable inlet guide vane has a first pitch, and wherein the second variable inlet guide vane has a second pitch different from the first pitch.

16. The apparatus of claim 12, wherein an aft portion of the first variable inlet guide vane contacts a forward portion of the second variable inlet guide vane.

17. The apparatus of claim 16, further including an actuator operatively coupled to the first variable inlet guide vane, wherein the actuator is to cause the first variable inlet guide vane to rotate, and wherein the second variable inlet guide vane rotates in an opposite direction from the first variable inlet guide vane when the actuator causes the first variable inlet guide vane to rotate.

18. An apparatus comprising: a first variable inlet guide vane including a first inner radial edge and a first outer radial edge, wherein a first distance from the first inner radial edge to the first outer radial edge defines a height of a flow path; and a second variable inlet guide vane downstream of the first variable inlet guide vane in the flow path, wherein the second variable inlet guide vane includes a second inner radial edge and a second outer radial edge, and wherein a second distance from the second inner radial edge to the second outer radial edge is less than the first distance.

19. The apparatus of claim 18, wherein the first variable inlet guide vane includes a first leading edge and a first trailing edge, and wherein the second variable inlet guide vane includes a second leading edge positioned downstream of the first leading edge and upstream of the first trailing edge in the flow path.

20. The apparatus of claim 18, wherein the first variable inlet guide vane includes a third outer radial edge positioned between the first inner radial edge and the second outer radial edge in a radial direction that defines the height of the flow path.
